Yi LIU

+86 183 2641 8258 | liuyi61@tongji.edu.cn | Ly403.github.io | 4800 Cao'an Road, Shanghai, China

Biography

Yi Liu is a master's student at ADMIS Lab, Tongji University, advised by Assoc. Prof. Wengen Li.

His research centers on **efficient and controllable generative models**, primarily diffusion and flow matching, across image restoration, image and video generation, and digital-human motion generation. His main focus is efficiency: making generation faster through few-step sampling, lightweight architectures, and hardware-aware acceleration. In parallel, he studies conditional control, from restoration guidance to history-conditioned, temporally coherent long-sequence modeling.

His longer-term goal is to **push generative models toward real-world impact** by accelerating generation at both the algorithmic and hardware-aware levels, making controllable generation reliable enough for real-time, interactive applications.

Education

Tongji University, **School of Computer Science and Technology** Shanghai, China
M.Eng. in Computer Science and Technology *Sept. 2024 — Mar. 2027 (expected)*
- Average Score: 93.88/100 | Advised by Assoc. Prof. Wengen Li | National Scholarship for Graduate Students

Tongji University, **College of Electronics and Information Engineering** Shanghai, China
B.Eng. in Computer Science and Technology *Sept. 2020 — Jul. 2024*
- Cumulative GPA: 4.65/5.0 (91.49/100) | Admitted to the master's program by recommendation (examination waived)

Publications

- [**NeurIPS 2026, Accepted**] **Yi Liu**, Xiangyue Zhang, Jianfang Li, Jia Ma, Yisheng He, and Jianqiang Ren, "Past as State, Present as Attention: Persistent-State Blockwise Flow Matching for Long-Horizon Co-Speech Motion Generation," 2026.
- [**CVPR 2025**] **Yi Liu**, Wengen Li, Jihong Guan, Shuigeng Zhou, and Yichao Zhang, "Effective Cloud Removal for Remote Sensing Images by an Improved Mean-Reverting Denoising Model with Elucidated Design Space," in *Proceedings of the Computer Vision and Pattern Recognition Conference (CVPR)*, 2025, pp. 17851–17861.
- [**arXiv**] **Yi Liu**, Jia Ma, Wengen Li, Jihong Guan, Shuigeng Zhou, and Yichao Zhang, "Disentangling Generation and Regression in Stochastic Interpolants for Controllable Image Restoration," *arXiv preprint* arXiv:2605.21381, 2026.
- [**arXiv**] Yifan Zhang, Qian Chen, **Yi Liu** (**co-first author**), Wengen Li, and Jihong Guan, "SADER: Structure-Aware Diffusion Framework with Deterministic Resampling for Multi-Temporal Remote Sensing Cloud Removal," *arXiv preprint* arXiv:2602.00536, 2026.

Research Experience

Controllable Image Restoration via Disentangled Stochastic Interpolants Tongji University, Shanghai, China
Project Leader, advised by Assoc. Prof. Wengen Li, ADMIS Lab *Jul. 2025 — Mar. 2026*
- Proposed **DiSI, a framework reconciling two opposing paradigms of image restoration** (fast, high-fidelity regression versus slower, realistic generation) by disentangling the stochastic interpolant into regression and generation components.
- Built **a single controllable model** that continuously transitions from faithful reconstruction to realistic generation at inference, validated across image deraining, image denoising, image super-resolution, and image inpainting.

Effective Cloud Removal by an Improved Mean-Reverting Denoising Model Tongji University, Shanghai, China
Project Leader, advised by Assoc. Prof. Wengen Li, ADMIS Lab *Oct. 2023 — Nov. 2024*
- Proposed **EMRDM, a mean-reverting diffusion framework** that diffuses directly between cloudy and cloud-free images instead of generating them from noise as standard diffusion does, preserving the inherent information in the cloudy input.
- Designed it modularly with a single unified and efficient network handling both mono-temporal and multi-temporal inputs, reaching **state-of-the-art cloud removal on public benchmarks**; independent first-author paper accepted to **CVPR 2025**.
- Released the full implementation of this project as open source (Ly403/EMRDM).

Efficient Reconstruction for Snapshot Compressive Imaging Westlake University, Hangzhou, China
Visiting Student, mentored by Prof. Xin Yuan, SCI Lab *Feb. 2024 — Jun. 2024*

- Independently adapted efficient architectures (e.g., **linear attention, state space models**) into SCI reconstruction for lower-cost inference, validated on benchmarks, and summarized 50+ papers in a technical report presented at group meetings.

SenseAir: AI-Driven Compressed Air System Management Platform Tongji University, Shanghai, China
Core Team Member *Mar. 2023 — Sept. 2023*

- Developed **SenseAir**, a digital-twin platform for compressed-air system operation, maintenance, and optimization; first inventor of a granted patent, with a registered software copyright.

Internship Experience

Inference Acceleration for Real-Time Interactive Audio-Video Avatar Generation ByteDance, Shanghai, China
Research Intern, GenAI & Ads Infra, Global Monetization Product and Technology *May 2026 — Present*

- Contributed to **a 3.6B live-streaming avatar model** that jointly generates synchronized audio and video for up to **~1 hour**.
- Enabled **interactive streaming with on-the-fly prompt switching during generation** by converting the non-autoregressive model to autoregressive generation via Self-Forcing, **preserving identity consistency** across prompt transitions.
- Achieved **real-time inference for the 3.6B model** by combining consistency-model training with distribution matching distillation (DMD), using a 14B teacher to reduce sampling from **50 to 4 steps**, together with hardware-aware kernel fusion.

Long-Horizon Co-Speech Motion Generation for 3D Digital Humans Alibaba Cloud, Hangzhou, China
Research Intern, Tongyi Lab (Multimodal Interaction) *Jun. 2025 — May 2026*

- **Identified, via quantitative analysis, a key asymmetry** overlooked by prior methods: in long-horizon co-speech motion generation, past motion encodes slow-varying style, while the current segment carries fast-changing dynamics.
- Following this, proposed **PASPA, a blockwise flow matching model** that summarizes history into a persistent state and models the current segment with blockwise attention, enabling long-sequence generation at constant memory.
- Achieved **state-of-the-art FGD** on public benchmarks; the resulting first-author paper was accepted at **NeurIPS 2026**.

Selected Honors and Awards

National Scholarship for Graduate Students, Ministry of Education of China *Dec. 2025*
Gold Prize, "Internet+" Innovation and Entrepreneurship Competition (Shanghai Regional) *Sept. 2023*
Third Prize, National University Student Contest on Energy Saving and Emission Reduction *Aug. 2023*
First Prize, Undergraduate Group of Asia and Pacific Mathematical Contest in Modeling *Mar. 2023*
Second Prize, Embedded System Design Invitational Contest (Intel Cup) *Aug. 2022*
Second and Third Prizes, Tongji University Outstanding Undergraduate Student Scholarship *2020 – 2024*

Skills

- **Programming**: Python (PyTorch, PyTorch Lightning) | C/C++ | Linux / Git | LaTeX
- **Languages**: Chinese: Native | English: IELTS 7.0 (Reading 8.5, Listening 7.5, Speaking 6.0, Writing 6.5); CET-6 557